

Experiment No: 4 Experiment Name: Demonstrate data visualization, histograms and multiple variable summaries

Aim:

To demonstrate various data visualization techniques including histograms, bar charts, scatter plots, and to generate multiple variable summaries for a given dataset.

Algorithm:

- Step 1: Import libraries (matplotlib, seaborn, pandas, numpy).
- Step 2: Load or generate a sample dataset.
- Step 3: Compute descriptive statistics using describe().
- Step 4: Plot a Histogram to show frequency distribution of a variable.
- Step 5: Plot a Bar Chart for categorical variable comparisons.
- Step 6: Plot a Scatter Plot to examine relationships between two variables.
- Step 7: Plot a Heatmap for correlation among all numeric variables.
- Step 8: Plot a Box Plot to visualize spread and outliers.
- Step 9: Display all plots and summaries.
- Step 10: Interpret the visualizations and draw conclusions.

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris

# Load Iris dataset
iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target_names[iris.target]

print("=== Dataset Summary ===")
print(df.describe())

fig, axes = plt.subplots(2, 2, figsize=(12, 10))

# Histogram
axes[0, 0].hist(df['sepal length (cm)'], bins=15, color='steelblue',
edgecolor='black')
axes[0, 0].set_title('Histogram - Sepal Length')
axes[0, 0].set_xlabel('Sepal Length (cm)')
axes[0, 0].set_ylabel('Frequency')
```

```

# Scatter Plot
for species, color in zip(iris.target_names, ['red', 'green', 'blue']):
    subset = df[df['species'] == species]
    axes[0, 1].scatter(subset['sepal length (cm)'], subset['sepal width
(cm)'],
                        label=species, color=color, alpha=0.7)
axes[0, 1].set_title('Scatter Plot - Sepal Length vs Width')
axes[0, 1].set_xlabel('Sepal Length'); axes[0, 1].set_ylabel('Sepal Width')
axes[0, 1].legend()

# Heatmap (Correlation)
corr = df.drop('species', axis=1).corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', ax=axes[1, 0], fmt='.2f')
axes[1, 0].set_title('Correlation Heatmap')

# Box Plot
df.drop('species', axis=1).boxplot(ax=axes[1, 1])
axes[1, 1].set_title('Box Plot - All Features')

plt.tight_layout()
plt.savefig('visualization_output.png')
plt.show()

```

Output:

=== Dataset Summary ===

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
max	7.900000	4.400000	6.900000	2.500000

Visualization plots saved as 'visualization_output.png'.

Result:

Data visualization techniques were successfully demonstrated. Histograms revealed the distribution of sepal length, scatter plots showed species separation, the correlation heatmap identified strong positive correlation between petal length and petal width, and box plots highlighted feature spreads and outliers.